

Technology #160

Implement a smart fuel pricing system that
adjusts fuel prices in real-time based on
factors such as road congestion,
emission-responsive charging, and the rebound
Engineering Technology-Transfer Dossier
effect.

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-09

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Implement a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, emission-responsive charging, and the rebound effect.

WHY DOES IT MATTER?

Documented problem 'fuel inefficiency' concerning car. User-submitted engineering problem (session input): "most fuel efficient car on earth". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 7 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in ml_data. 7-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Conduct a controlled experiment in a major city with a large fleet of vehicles, comparing the effects of the smart fuel pricing system to.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: held-out + temporal validation splits. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data; empirical risk on a non-representative sample says nothing about deployment risk (distribution shift); utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured; ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.; ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.; ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.; Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII.); Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Dynamic pricing and optimization of fuel consumption based on real-time data analytics and machine learning algorithms.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'fuel inefficiency' concerning car. User-submitted engineering problem (session input): "most fuel efficient car on earth". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: ml_data

Disciplines: data engineering, machine learning, software engineering, statistics

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: Implement a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, emission-responsive charging, and the rebound effect.

Subsystems:

?: data ingestion and labeling pipeline - ?

?: feature / representation layer - ?

?: model + uncertainty quantification - ?

?: deployment target + monitoring loop - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Implement a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, emission-responsive charging, and the rebound effect.

Key Physics: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: APPLICABLE; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: $P(x|y)$.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: APPLICABLE; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: $n, f(x_i)$.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: CONDITIONAL; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: NOT ESTABLISHED.

Assumptions:

- priors and likelihoods are MODEL_DERIVED until validated on deployment data
- empirical risk on a non-representative sample says nothing about deployment risk (distribution shift)
- utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured
- ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.
- ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.
- ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.
- Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).

- Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'fuel inefficiency', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'adversarial dimension: weak_transfer', 'basis': '"COMPUTED (attack engine verdict) + domain-registry mechanism cross-reference: \'overfitting to source-site artifacts\' (E15-D; the adversarial finding describes a physical phenomenon the domain registry documents for this invention class)"}'
- {'regime': 'distribution shift degrading accuracy post deployment', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'label noise propagating to biased predictions', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'overfitting to source-site artifacts', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'calibration drift invalidating decision thresholds', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'feedback-loop data contamination', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	fuel efficiency	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Dynamic pricing and optimization of fuel consumption based on real-time data and machine learning algorithms	MODELED
DI-003	Expected effect target	Decreased gasoline consumption by 10% and improved overall fuel efficiency of vehicle fleet by 15%.	MODELED
DI-004	Domain design input pattern: decision applicability	5120 tokens ['decision']	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: available applicability	NOT ESTABLISHED (candidate pattern; confirm against engineering data)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: latent applicability	NOT ESTABLISHED (candidate pattern; confirm against engineering data)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidates	For example, the EPA's 2025 standards for greenhouse gas emissions are never inferred	ENGINEERING_PROPOSED
DI-008	Custodied observation: Emissions Response / Change in Emissions	45071625, starting to improve 449889907857074980bbdc6f0b6b555e38540	
DI-009	Custodied observation: Dissecting data principles	in article A1094785, regression analysis used 3375048036402196458188b14dccb5fb5c03	
DI-010	Custodied observation: Within-day variation in the reported effect	for 41503287, meta-analysis 451889117186706e355d30179d60129ed90760dbb6a	
DI-011	Custodied observation: Energetic and Ecologic Aspects of Regio	42005600 Application has the Eff 7041588076e7fa020a6509ef3032bb2032dade9	
DI-012	Custodied observation: Energetic and Ecologic Aspects of Regio	42006176, correlation has a R 55619059649354500114f0203f64994426108bd87	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	data ingestion and labeling pipeline realizing Implementation of a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	CONCEPTUAL	data that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns
DO-002	feature / representation layer realizing Implementation of a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	CONCEPTUAL	data that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns
DO-003	model + uncertainty quantification realizing Implementation of a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	CONCEPTUAL	data that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns
DO-004	deployment target + monitoring loop realizing Implementation of a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	CONCEPTUAL	data that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns
DO-005	data ingestion and labeling pipeline interfaces with feature / representation layer in a smart fuel pricing system	CONCEPTUAL	signature / utility function (time offsets, temperature)
DO-006	feature / representation layer interfaces with model + uncertainty quantification in a smart fuel pricing system	CONCEPTUAL	signature / utility function (time offsets, temperature)
DO-007	model + uncertainty quantification interfaces with deployment target + monitoring loop in a smart fuel pricing system	CONCEPTUAL	signature / utility function (time offsets, temperature)
DO-008	Device boundaries: patient/fluid inlet and outlet termination of control/data streams by selection and definition of pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	CONCEPTUAL	control/data streams by selection and definition of pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns
DO-009	Flow/force/signal path geometry realizing the mechanism of a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	UNKNOWN	algorithm, data inputs, feedback
DO-010	Operating range for training data volume and provenance of the mechanism of a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	UNKNOWN	algorithm, data inputs, feedback
DO-011	Operating range for label quality rate within which the mechanism holds its state bounds for label quality rate; verification of range endpoints	UNKNOWN	algorithm, data inputs, feedback
DO-012	Operating range for target operating point (precision/recall) within which the mechanism holds its state bounds for target operating point (precision/recall); verification of range endpoints	UNKNOWN	algorithm, data inputs, feedback
DO-013	Operating range for calibration error budget within which the mechanism holds its state bounds for calibration error budget; verification of range endpoints	UNKNOWN	algorithm, data inputs, feedback
DO-014	Operating range for inference latency ceiling within which the mechanism holds its state bounds for inference latency ceiling; verification of range endpoints	UNKNOWN	algorithm, data inputs, feedback
DO-015	Operating range for distribution-shift monitoring thresholds within which the mechanism holds its state bounds for distribution-shift monitoring thresholds; verification of range endpoints	UNKNOWN	algorithm, data inputs, feedback
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism of a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, weather, and traffic patterns	PROPOSED	sensing, prediction, actuation, pricing system
DO-017	Information pipeline: ingestion -> feature/representation -> model -> uncertainty quantification -> deployment target -> monitoring loop	PROPOSED	ingestion, feature/representation, model, uncertainty quantification, deployment target, monitoring loop
DO-018	Fixture enabling: held-out + temporal validation splits -> model selection	UNKNOWN	held-out, temporal validation splits, model selection
DO-019	Fixture enabling: calibration and decision-threshold analysis -> model selection	UNKNOWN	calibration, decision-threshold analysis, model selection
DO-020	Fixture enabling: failure-mode stress tests (distribution shift) -> model selection	UNKNOWN	failure-mode stress tests, distribution shift, model selection

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: held-out + temporal split evaluation	UNKNOWN	Unknown; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
training data volume and quality	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
label quality rate	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
target operating point (precision)	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
calibration error budget	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
inference latency ceiling	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
distribution-shift monitoring	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
fuel inefficiency	Fails by fuel inefficiency: the engine fails to meet its design requirements	design failure mode effects analysis (DFMEA) & prototyping	bench testing & engine test cell	operating test cell candidate failure mode
adversarial dimensionality	Fails by adversarial dimensionality: model overfits to training data	multisite transfer data; model ensembles; adversarial augmentation	adversarial attack benchmarks	adversarial attack signatures; model failure mode
distribution shift degradation	Fails by distribution shift: deployment and testing environments differ	domain adaptation; synthetic data; robustness training	domain shift stress tests	deployment candidate data fatigue mode
label noise propagation	Fails by label noise: mispropagation of biased predictions	robust loss functions; self-supervised learning; active learning	noise injection simulations	noise injection candidate failure mode
overfitting to source-sites	Fails by overfitting to source-sites: model lacks generalization	data augmentation; cross-site validation	cross-site validation	deployment candidate failure mode
calibration drift invalidation	Fails by calibration drift: invalidating decision thresholds	dynamic thresholding; probabilistic models	calibration benchmarks	calibration drift candidate failure mode
feedback-loop data corruption	Fails by feedback-loop data corruption: external data poisoning	data sanitization; anomaly detection	data integrity audits	data poisoning candidate failure mode

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
fuel inefficiency	Fails by fuel inefficiency: the device fails to maintain optimal fuel consumption	design optimization, NOT ESTABLISHED	fuel inefficiency reported as condition at failure the device (Annex A)
adversarial dimension: weak transfer	Fails by adversarial dimension: weak transfer of learned knowledge to new tasks	transfer learning, data augmentation, adversarial training	adversarial dimension: weak transfer (Annex A)
distribution shift degrading accuracy	Fails by distribution shift degrading accuracy	domain adaptation, robust training	distribution shift degrading accuracy (Annex A)
label noise propagating to model	Fails by label noise propagating to model	robust loss functions, data cleaning	label noise propagating to model (Annex A)
overfitting to source-site artifacts	Fails by overfitting to source-site artifacts	cross-validation, regularization	overfitting to source-site artifacts (Annex A)
calibration drift invalidating decisions	Fails by calibration drift invalidating decisions	regular calibration, monitoring	calibration drift invalidating decisions (Annex A)
feedback-loop data contamination	Fails by feedback-loop data contamination	data sanitization, monitoring	feedback-loop data contamination (Annex A)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Conduct a controlled experiment in a simulator	Simulation with realistic vehicle models	Controlled experiment results	Controlled experiment results (Annex A)
VF-002	run killer experiment: falsification_test	run killer experiment (Falsification Test)	run killer experiment results	run killer experiment results (Annex A)
VF-003	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria (Annex A)
VF-004	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria	held-out + temporal validation splits. Acceptance criteria (Annex A)
VF-005	failure-mode stress tests (distribution shift)	failure-mode stress tests (distribution shift)	failure-mode stress tests (distribution shift)	failure-mode stress tests (distribution shift) (Annex A)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed	Validation results (Annex A)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
NOT_APPLICABLE — data only invention: no physical materials exist to select; hardware dependencies are recorded under design	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
NOT	ESTABLISHED (no domain data)	NOT	ESTABLISHED (no domain data)				

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- MLOps CI for data and model: domain registry manufacturing route - candidate only; no process qualification exists
- versioned artifact registry with rollback: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter training data volume and provenance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter label quality rate [y_i -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter target operating point (precision/recall) [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter calibration error budget [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter inference latency ceiling [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter distribution-shift monitoring thresholds [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article implementation of the proposed physical verification (to be quoted)	per method	conduct a controlled experiment	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Conduct a controlled experiment for VF-001 verification	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	held-out + temporal validation for VF-003 verification	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	held-out + temporal validation for VF-004 verification	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	failure-mode stress test for VF-005 verification	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.